Order Proboscidea, Family Elephantidae

Elephants (*Elephas maximus*, *Loxodonta africana*) are large (1000-7500 kg), diurnal, terrestrial herbivores in Africa and Asia. Males are larger than females (281a, 6292) and have a temporal gland which is active during musth, a period of male sexual activity (1239a, 1484, 1928, 2040a, 2058, 2856, 2936, 4245, 5235, 7262, 8660, 8661, 9659). Flehmen-like behavior occurs (8939).

The labyrinthine, deciduate, zonary placenta has been described as hemochorial, endotheliochorial, epitheliochorial, and vasochorial, and implants into a bicornuate or duplex uterus (241, 434, 706, 1040, 1715, 2041, 2391, 7262, 7563, 8395, 8396, 8397, 9972, 10310, 11485). Placentophagia is occasional (2596, 3063, 5762, 6412, 6681, 9972). Estrus does not occur postpartum (4320, 6119). A breech birth is reported (6733a). Neonates are precocial with open eyes and developed body hair (330, 2596, 5762, 6681, 7751, 9972, 10158). Females have 2 thoracic mammae located between the forelegs (331, 1715, 7262). Males have abdominal testes, seminal vesicles, and a sigmoid penis (3823, 4322, 5427, 9665, 9941, 11479).

Details of anatomy (540, 541, 4321, 4327, 4937, 6282, 6283, 8396, 9665, 9935a, 10135, 11484, 11485, 11487), endocrinology (1239, 1592, 1731, 2412, 4327, 4701, 4702, 4869, 5235, 5549, 6733a, 7111, 8048, 8602, 8607, 8894, 8940, 9935a, 9936, 10116, 10691a), reproductive cyclicity (9350), sperm maturation (2264, 4937, 5357, 5426, 5427, 5429, 5430), copulation (2936, 2941, 5233, 5762), birth (6187), suckling (6351), and milk composition (7039, 8401) are available.

NEONATAL MASS

Elephas maximus	r79.4-96.8 kg; n3	3341
	90 kg, n1	308
	~90 kg	7751
	~90.7 kg; 2 d	8501
	91 kg	8498
	m: 91.6 kg; r50-150; n14	2597
	92.5 kg; r83-102; n2	6681
	f: 93 kg; n1	11047
	97 kg; n1	1715
	est 100 kg	3063
	~100 kg	10491
	101 kg; n1	1283
	f: 101.2 kg; r77-127	2597
	104.35 kg; r70-133; n10	9065
	107 kg; r50-150; n25	2596
	114.25 kg; r106.5-122; n2	2597
	138 kg; n1, 1.5 d	6733a
Loxodonta africana	73 kg; term emb	331
	78.5 kg; r78-79; n2	1485
	89.8 kg; r84.1-95.5; n2 term	9310
	f: 90-100 kg	9972
	94.98 kg; r80.0-113	9310
	113 kg	9065
	113 kg	6187
	~118 kg	4319
	120 kg	6290, 6292
	m: ~120 kg	

LITTER SIZE

Elephas maximus	1	6119
	mode 1; r1-2	8501
	1 stillborn; n1 lt	706
	1 yg; n1 lt	6032
	1 yg; n1 lt	11047
	twins; n1 set: 1 m, 1 f	12018
Loxodonta africana	1	8514
	1	331
	1	11253
	1 cl	4327
	1 egg fertilized	8394
	1; 2 occasional	6292
	1; 2 rare	9972
	1; n1	6187
	1; r1-2; 0.02% twins; n6000 lt	7204

	<1% twins	10135
	1.01 emb; r1-2; 0.6% twins; n351 f	10163, 10165
	2 emb; n1 f	9310
	2 emb; n1 f	9801
SEXUAL MATURITY		
Elephas maximus	f: 1st conceive: 6-7 yr	2941
	f: 1st mate: ~8 yr	6119
	f: 1st birth: ~10 yr	6119
	f: 1st birth: 13, 14-16 yr; n2	8501
	f: 1st birth: 15-16 yr	3341
	m: 1st sire: 7 yr	2941
	m: 1st sire: 14-15 yr	3341
	f/m: min 9 yr	2597
Loxodonta africana	f: follicles large: 14.42 yr; ±1.17	6290, 6292
	f: 1st mate: 13-14 yr	6282
	f: 1st ovulate: 11 yr	11731
	f: 1st ovulate: 12-13 yr	5698
	f: 1st ovulate: 14 yr	4320
	f: 1st cl: 12-13 yr	9901
	f: 1st cl: 17.8 yr; r12-22	6292
	f: 1st conceive: 7 yr	5202
	f: 1st conceive: 12-14 yr	6349
	f: 1st preg: min 9 yr; 75% 12 yr; few data	10163
	f: 1st preg: 7-15 yr	10134
	f: preg/lact: ~15 yr	7469
	f: 1st birth: 8-12 yr	8395
	f: 1st birth: ~9 yr	1485
	f: ~11 yr	10158
	f: 13-15 yr	4319
	f: 15-20 yr	6057
	m: sperm: 7-8 yr; abundant 9 yr	5358
	m: sperm: 8-12 yr	8395
	m: sperm: 10 yr; r3-15	5356
	m: spermatogenesis: 13 yr	9901
	m: spermatogenesis: 14 yr	5698
	m: spermatogenesis: 14.43 yr; ±0.76	6290, 6292
	m: spermatogenesis: 15 yr	4320
	m: testes large: 6-8 yr	4322
	m: tubule diam increases: 15 yr	4322
	m: 1st repro: 17.2 yr; r10-17; sperm 2-3 yr earlier	6292
	m: ~10 yr	10158
	m: 15-20 yr	6057
	m: 1800 kg	6291
	f/m: 1st mate: 7 yr	1485
	f/m: 8-13 yr	9972
	f/m: 12-13 yr	8514
CYCLE LENGTH		
Elephas maximus	estrus: mode 4 d; r2-8	2936, 2941, 5233
	follicular phase: 4.2 wk; ±0.5	8607
	follicular phase: 6.8 wk; ±1.1	4702
	luteal phase: 10.5 wk; ±0.3	4701
	luteal phase: 10.6 wk; ±0.6	8607
	luteal phase: 10.7 wk; ±0.3	4702
	17.7 d; r18-26; n3 cycles, 2 f	8894
	22 d; r18-27; n6 cycles, 6 f	2936, 2941, 5233
	3.5 mo; behavior	8939
	14.7 wk; se0.5; n10 cycles, 2 f	8607
	16.1 wk; sd2.1; n5 cycles, 1 f, between progesterone peaks	6733a
	16.3 wk; n15 cycles, 6 f; behavior	4701
	16.6 wk; sd1.6; n5 cycles, 1 f, between estrogen peaks	6733a
	16.8 wk; se0.6; n14 cycles, 2 f, between progesterone peaks	10691a
	18 wk; n14 cycles, 7 f	4702

Loxodonta africana	estrus: 1-2 d	5762
	estrus: 2-10 d; n7 f	7560
	follicular phase: 5.9 wk; ±0.6	8607
	luteal phase: 10.0 wk; ±0.3	8607
	16 d; n1 f; smears	11487
	13.3 wk; ±1.3; n11 cycles, 3 f	1239
	15.9 wk; se0.6; n4 cycles, 25 f	8607
GESTATION		
Elephas maximus	17 mo 17 d-24 mo 13 d	1453
	18-22 mo	308
	18-23 mo	8501
	mode 19 mo; r18-22	8498
	20 mo 18 d	2058
	20 mo 20 d	1355
	20 mo 21 d	4582
	21 mo	6119
	21 mo 18 d; n1	102
	630-656 d; last vs first mating to birth	1715
	630.3 d; r624-637; n3	3063
	634.5 d; r634-635; n2	6681
	643.7 d; r614-668; n15	2597
	646 d	11047
	666 d; n1	6032
	644 d; r628-668; n3	2596
	~25 mo; n1	10491
Loxodonta africana	20-20.5 mo; stated; 20-24 mo; field data	11697
	21-22 mo	5762
	649-661 d; n1; last vs first mating to birth	6187
	~22 mo	4320
	22 mo	5202
LACTATION		
Elephas maximus	solid food: est 3 mo	7751
	solid food: 4-5 mo	6681
	wean: ≤18 mo	9946
Loxodonta africana	solid food: 3 mo	6349
	dependent on milk: 24 mo	6349, 6351
	nurse: 2 yr	9972
	nurse: 4-8 yr	6292
	wean: 6-18 mo	5762
	wean: 36 mo	6349
	wean: est 4.8 yr	10134
INTERLITTER INTERVAL		
Elephas maximus	min 2 yr	3658
	2-2.5 yr; n1	12018
	4 yr; r3-5 yr	6119
Loxodonta africana	2-2.5 yr	9972
	2.5-9 yr	5762
	3.3-3.7 yr	9901
	3.35 yr	4320
	3.5 yr	5698
	3.5 yr	6412
	3.5 yr; good habitat; 7.6 yr; poor habitat	7469
	3.9 yr; sd1.1; r2.2-5.3; n15	5202
	4 yr	11731
	4 yr	8514
	4 yr	4319
	~4 yr	8395
	4.5 yr	10165
	5 yr	7560
	5.6 yr; r4.8-6.8	6290, 6292
	6.7 yr	6282
	8 yr	10134
	9.1 yr; r2.6-13.5	6292

SEASONALITY

Elephas maximus	mate: yr rd; India	2058
	repro: yr rd; Sri Lanka	8501
	repro: yr rd, peak Mar-Oct, Dec, low Jan-Feb, Sept-Nov; zoo	2597
Loxodonta africana	hormones: high yr rd; Uganda	7111
	mate: Jan-Feb; Angola	11697
	conceive: Oct-July, peak Jan-Mar, 88% Nov-Apr; Zambia	4318, 4320
	conceive: Nov-Apr 70%; S Africa	10165
	conceive: Nov-Apr 88%; Zimbabwe	11731
	conceive: Nov-Apr; Zimbabwe	5698
	birth: Apr-June; Zaire	11253
	birth: est Oct-Dec; Okawango, Angola	11697
	birth: yr rd, peaks Feb, July, Nov; Bunyory, Uganda	6292
	birth: yr rd; Zimbabwe	331
	birth: yr rd; Uganda	8394
	birth: yr rd; S Africa	3145
	birth: yr rd, peak before peak rain; central, e, se Africa	5762
	repro: yr rd; S Africa	10158